

Solution: Problem Set - 3

Question 1:

Consider $U = \{(m, n) : m, n \in \mathbb{Z}\}$, where $\mathbb{Z}$ is the set of *integers*.

Let $v := (m_1, n_1)$ and $w := (m_2, n_2)$ are the sets $v, w \in U$. Then, U is closed under addition and under taking additive inverses, since we have

$$\begin{aligned} v + w &= (m_1 + m_2, n_1 + n_2) \in U \\ -v &= (-m_1, -n_1) \in U \quad -w = (-m_2, -n_2) \in U. \end{aligned}$$

But, U is not closed under scalar multiplication, as αv may not be in set U e.g. for $v = (1, 3) \Rightarrow \frac{1}{2} \cdot v = (\frac{1}{2}, \frac{3}{2}) \notin U$. Hence, U is not a subspace in $\mathbb{R}^2$.

Question 2:

Consider $U = \{(x, y) \in \mathbb{R}^2 : x = 0 \text{ OR } y = 0\}$, where $\mathbb{R}$ is the set of *real numbers*.

Let $v := (x_1, 0)$ and $w := (0, y_1)$, implying $v, w \in U$. Then, U is closed under scalar multiplication, since we have

$$\begin{aligned} \alpha \cdot v &= (\alpha \cdot x_1, 0) \in U \\ \alpha \cdot w &= (0, \alpha \cdot y_1) \in U. \end{aligned}$$

But, U is not closed under addition, e.g. for $v = (1, 0) \in U, w = (0, 1) \in U \Rightarrow v + w = (1, 0) + (0, 1) = (1, 1) \notin U$. Hence, U is not a subspace in $\mathbb{R}^2$.

Question 3:

Suppose that $\{\underline{v}_1 + \underline{w}, \dots, \underline{v}_n + \underline{w}\}$ is linearly dependent. Then, there exist $a_1, \dots, a_n$ not all equal to zero, such that

$$\begin{aligned} a_1(\underline{v}_1 + \underline{w}) + \dots + a_n(\underline{v}_n + \underline{w}) &= 0 \\ a_1\underline{v}_1 + \dots + a_n\underline{v}_n + (a_1 + \dots + a_n)\underline{w} &= 0 \\ a_1\underline{v}_1 + \dots + a_n\underline{v}_n &= -(a_1 + \dots + a_n)\underline{w}. \end{aligned}$$

Now, if $b = -(a_1 + \dots + a_n) = 0$, then $a_1\underline{v}_1 + \dots + a_n\underline{v}_n = 0$ which is a contradiction as not all the a_i s are zero but $\underline{v}_i$ s are linearly independent. Hence, $b \neq 0$, and thus

$$\underline{w} = \frac{a_1}{b}\underline{v}_1 + \dots + \frac{a_n}{b}\underline{v}_n.$$

which implies $\underline{w} \in \text{span}(\underline{v}_1, \dots, \underline{v}_n)$.

Question 4:

We can rewrite the equations as

$$\begin{aligned} U &= \{(3u_2, u_2, 7u_4, u_4, u_5) | u_2, u_4, u_5 \in \mathbb{R}\} \\ &= \{u_2(3, 1, 0, 0, 0) + u_4(0, 0, 7, 1, 0) + u_5(0, 0, 0, 0, 1)\} \\ &= \text{span}((3, 1, 0, 0, 0), (0, 0, 7, 1, 0), (0, 0, 0, 0, 1)). \end{aligned}$$

To prove that these elements are in fact the basis, we must show these are linearly independent.

$$a(3, 1, 0, 0, 0) + b(0, 0, 7, 1, 0) + c(0, 0, 0, 0, 1) = (0, 0, 0, 0, 0).$$

Then, $(3a, a, 7b, b, c) = (0, 0, 0, 0, 0)$ implies $a = b = c = 0$. Hence, the set is linearly independent and spans the U , is the basis of U .

Question 5:

The natural basis for the question is $\{x^3, x^2, x, 1\}$. We need to find the set without polynomial 0 and 2 degree. If v_1, v_2, v_3, v_4 are linearly independent, then so are the $v_1, v_1 + v_2, v_1 + v_2 + v_3, v_1 + v_2 + v_3 + v_4$. Indeed, let

$$0 = a_1v_1 + a_2(v_1 + v_2) + a_3(v_1 + v_2 + v_3) + a_4(v_1 + v_2 + v_3 + v_4).$$

Then

$$0 = (a_1 + a_2 + a_3 + a_4)v_1 + (a_2 + a_3 + a_4)v_2 + (a_3 + a_4)v_3 + (a_4)v_4.$$

As v_1, v_2, v_3, v_4 are linearly independent, we have

$$(a_1 + a_2 + a_3 + a_4) = 0 \quad (a_2 + a_3 + a_4) = 0 \quad (a_3 + a_4) = 0 \quad (a_4) = 0.$$

which implies that $a_1 = a_2 = a_3 = a_4 = 0$. Then,

$$\{x^3, x^3 + x^2, x^3 + x^2 + x, x^3 + x^2 + x + 1\}.$$

is a set containing all 3 degree polynomial coefficients, which is L.I. Since $\dim(\mathcal{P}_3) = 4$, these vectors span $\mathcal{P}_3$ as well. Hence, it is a basis.

Question 6:

Assume that there exists S , a subspace of V , such that $\{\underline{v}_1, \dots, \underline{v}_k\} \subset S$ but $\text{span}(\underline{v}_1, \dots, \underline{v}_k) \not\subset S$.

Therefore, $\exists \underline{w} \in \text{span}(\underline{v}_1, \dots, \underline{v}_k)$ such that $\underline{w} \notin S$. Since, $\underline{w} \in \text{span}(\underline{v}_1, \dots, \underline{v}_k) \Rightarrow \underline{w} = \sum_{i=1}^k \alpha_i \underline{v}_i$. Since, $\underline{v}_i \in S$ and S is a subspace, any linear combination of $\underline{v}_i$ s is also in S , we have $\sum_{i=1}^k \alpha_i \underline{v}_i \in S \Rightarrow \underline{w} \in S$, which is contradiction. Hence, $\text{span}(\underline{v}_1, \dots, \underline{v}_k)$ is the smallest subspace that contains $\{\underline{v}_1, \dots, \underline{v}_k\}$.

OR

We need to show that any subspace of V that contains $\{\underline{v}_1, \dots, \underline{v}_k\}$ also contains $\text{span}(\underline{v}_1, \dots, \underline{v}_k)$. Accordingly, let W be a subspace of V that contains $\{\underline{v}_1, \dots, \underline{v}_k\}$. Then since W is closed under addition and scalar multiplication, it contains every linear combination $\{c_1\underline{v}_1, \dots, c_k\underline{v}_k\}$ of $\{\underline{v}_1, \dots, \underline{v}_k\}$. Therefore, $\text{span}(\underline{v}_1, \dots, \underline{v}_k)$ is contained in W .